

CyberVerse AI - Technical Architecture Document

2.1 High-Level Diagram

2.2 Data Model (key columns)

Table	Key columns
users	email, hashed_password, full_name, skill_level, certifications[], weak_topics[], learning_goals, xp, streak_days, last_active_date, is_pro, is_premium, created_at
mentor_messages	user_id, role, content
daily_bundles	user_id, date, content(JSON)
resume_reviews	user_id, role, review(JSON)
interview_sessions	user_id, role, status(active/completed), overall_score
interview_turns	session_id, speaker, content, feedback
achievements	user_id, type, title, date

2.3 API Surface (principal endpoints)

Method	Path	Purpose
POST	/api/auth/signup, /login, /forgot-password, /reset-password	Auth
GET	/api/profile/me	Profile incl. is_pro
POST	/api/mentor/chat	Context-aware mentor
GET	/api/daily/today	Daily bundle
POST	/api/resume/review	Resume analysis
POST	/api/interview/start, /{id}/respond	Interview lifecycle
GET	/api/interview/active	Session resume
POST	/api/webhooks/razorpay	HMAC-verified upgrade
GET	/api/admin/stats	Founder metrics (allowlist)

2.4 Deployment and Ops

GitHub push -> Render build -> Alembic migrations -> idempotent runtime ALTER safety net -> Gunicorn serve. Config via pydantic-settings.

2.5 Environment Variables

DATABASE_URL, JWT_SECRET, ANTHROPIC_API_KEY, RESEND_API_KEY,
RESEND_FROM_EMAIL, RAZORPAY_WEBHOOK_SECRET, ADMIN_EMAIL(S),
APP_BASE_URL, RATE_LIMIT_

2.6 Scaling Path

Stateless API -> horizontal workers; extract cron to external scheduler/queue; Redis for rate-limit + cache; S3/R2 for resume files; CDN for static assets; read replicas when needed.